1.Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-

making.

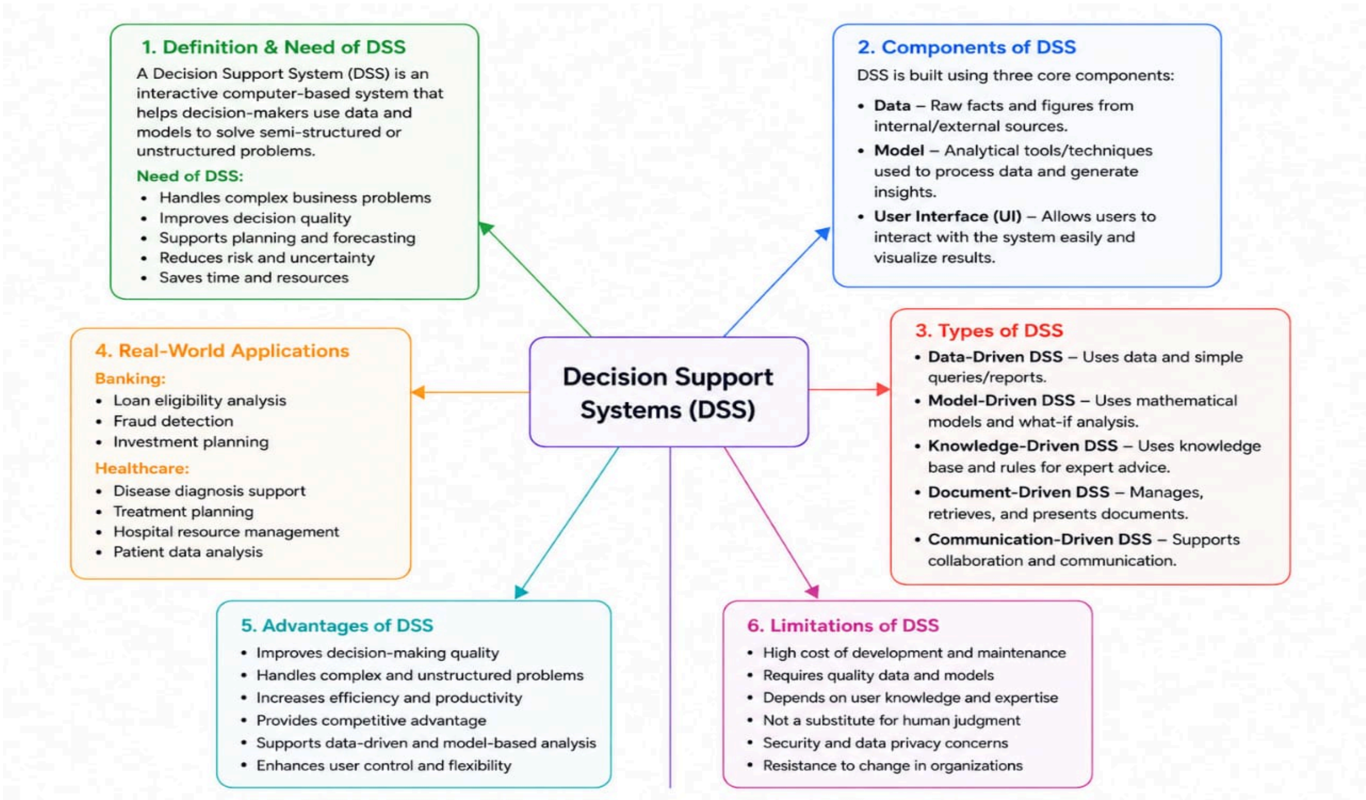
Core components: Data, Model, User Interface

1. Types of DSS

2. Flow: Data → Analysis → Decision

3. Role in business decision-making

4. Real-world example linkage



2. Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support

Systems (OLAP).

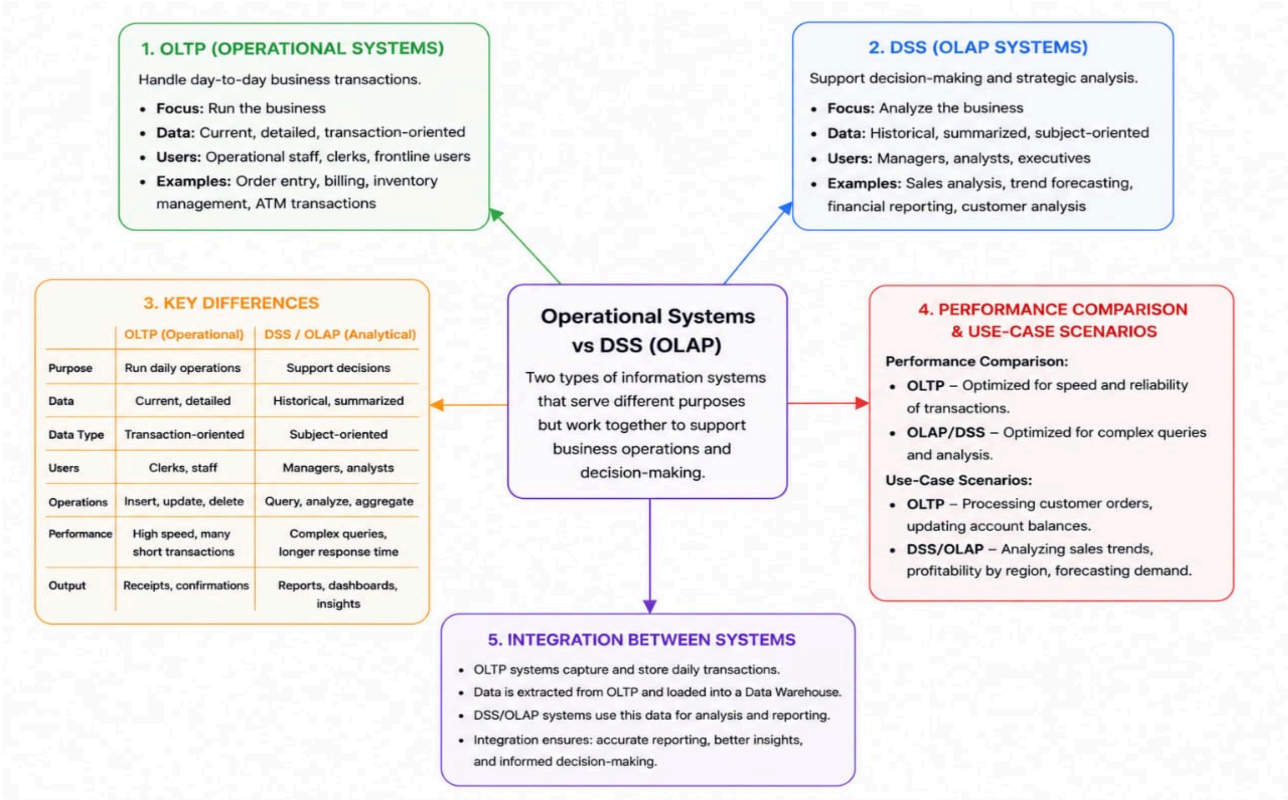
1. Definitions of OLTP and DSS

2. Differences (data type, purpose, users, processing)

3. Similarities and interactions

4. Real-time vs historical data

5. Example systems (banking vs analytics)



3. Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

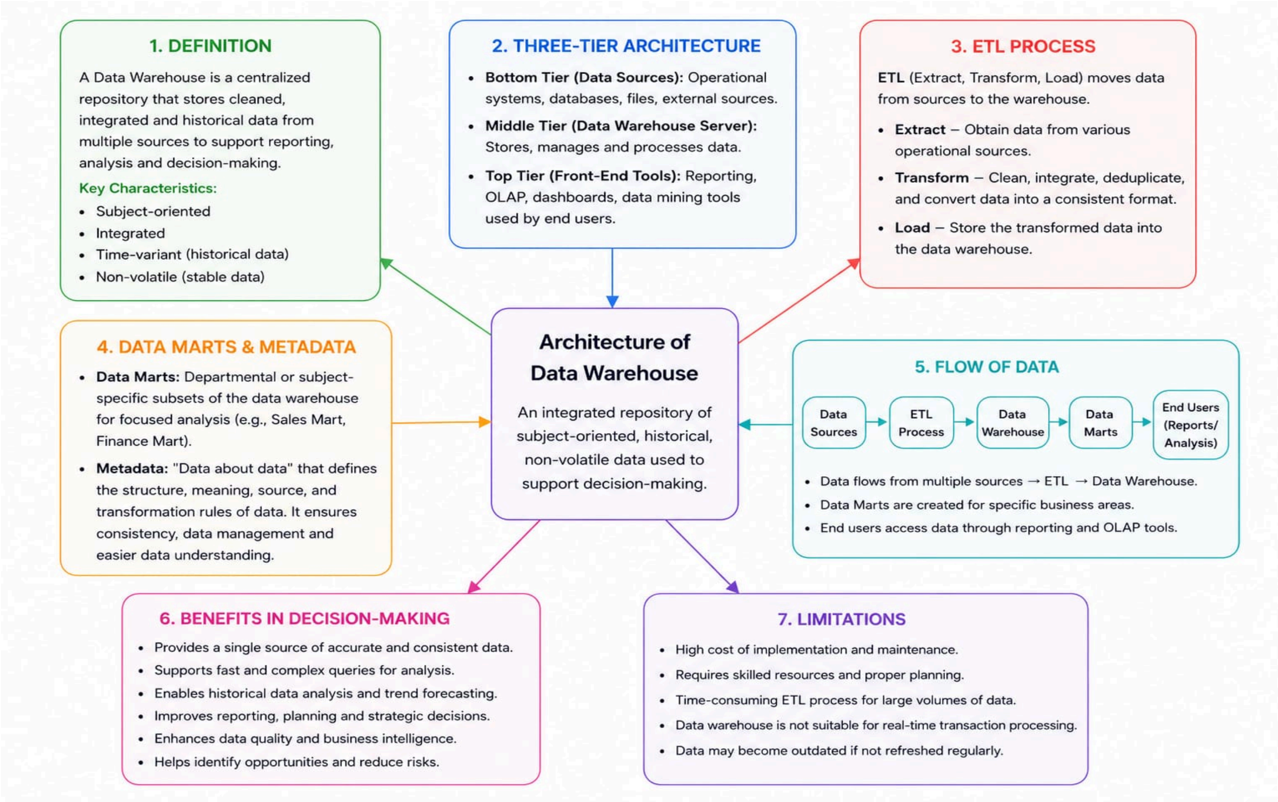
1. Data Sources → ETL → Data Warehouse → Data Marts

2. Metadata layer

3. Front-end tools (reporting, OLAP)

4. Data flow direction

5. Role of each layer



4. Concept Map 4: OLAP, Data Cubes & Dimensional Modelling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modelling techniques.

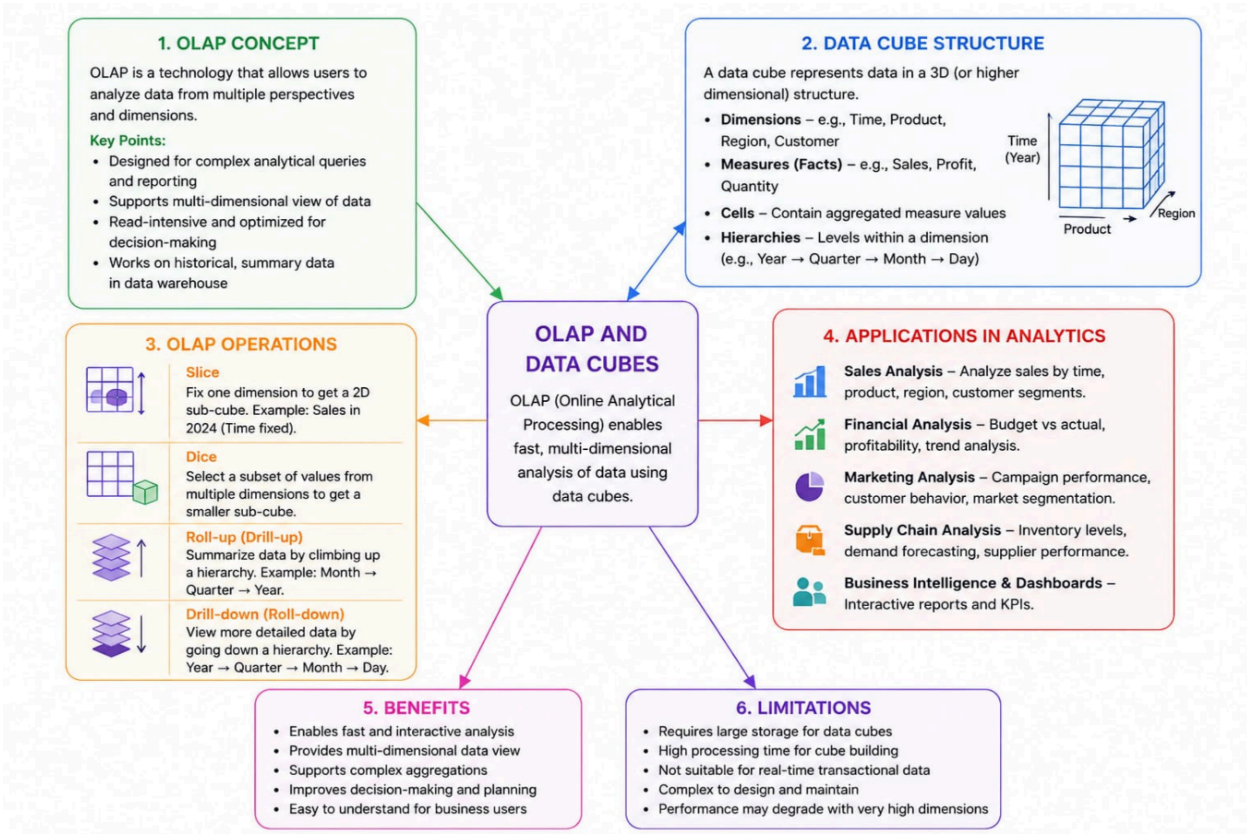
1. Data cube structure (dimensions, measures)

2. OLAP operations (slice, dice, roll-up, drill-down)

3. Star schema and snowflake schema

4. Fact and dimension tables

5. Analytical usage



5. Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for

reporting and OLAP applications.

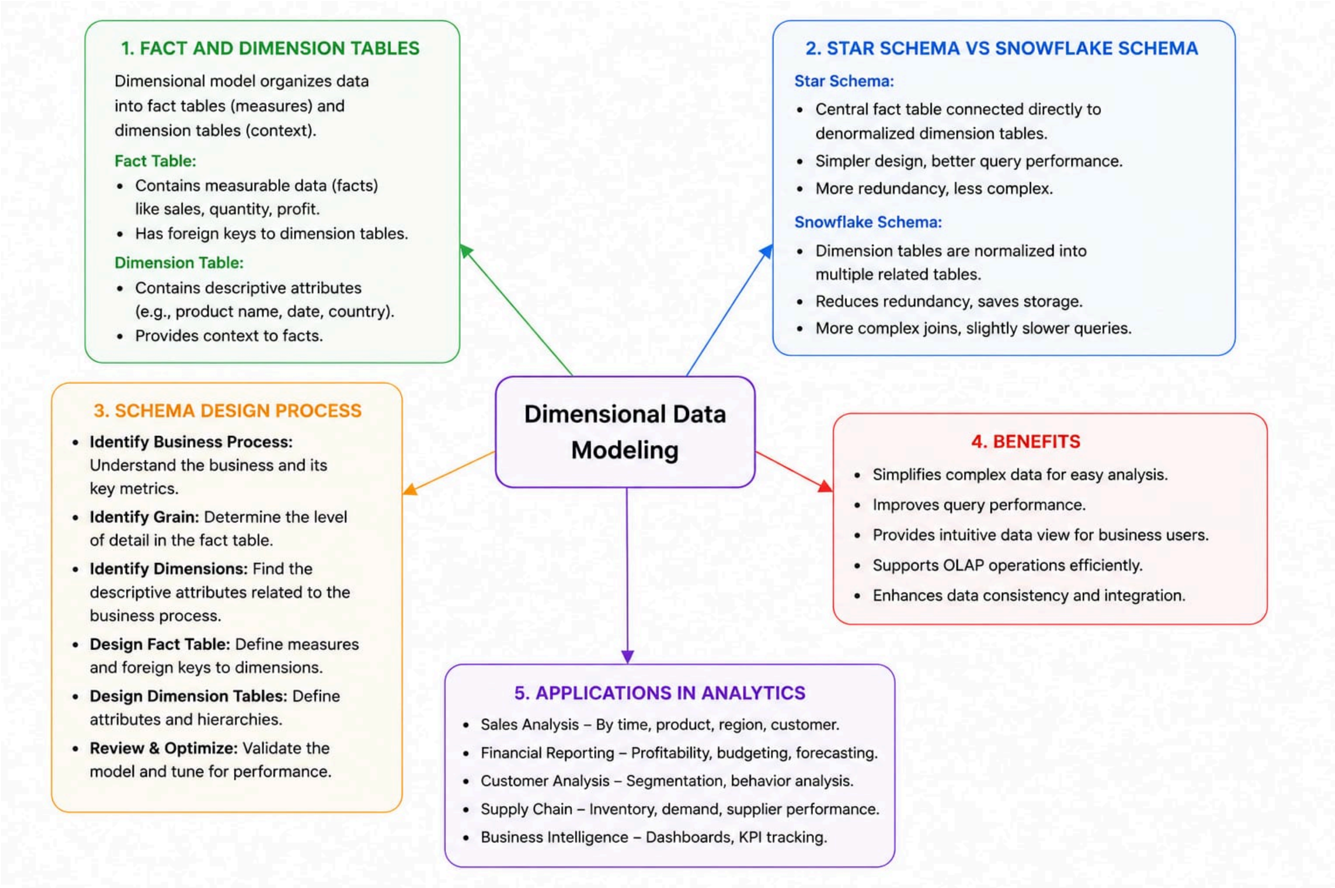
1. Query processing stages (Parsing, Optimization, Execution)

2. Role of BI tools

3. KNIME workflow structure

4. Data → Query → Visualization pipeline

5. OLAP application integration



6: Query Processing & BI Tools

Task description

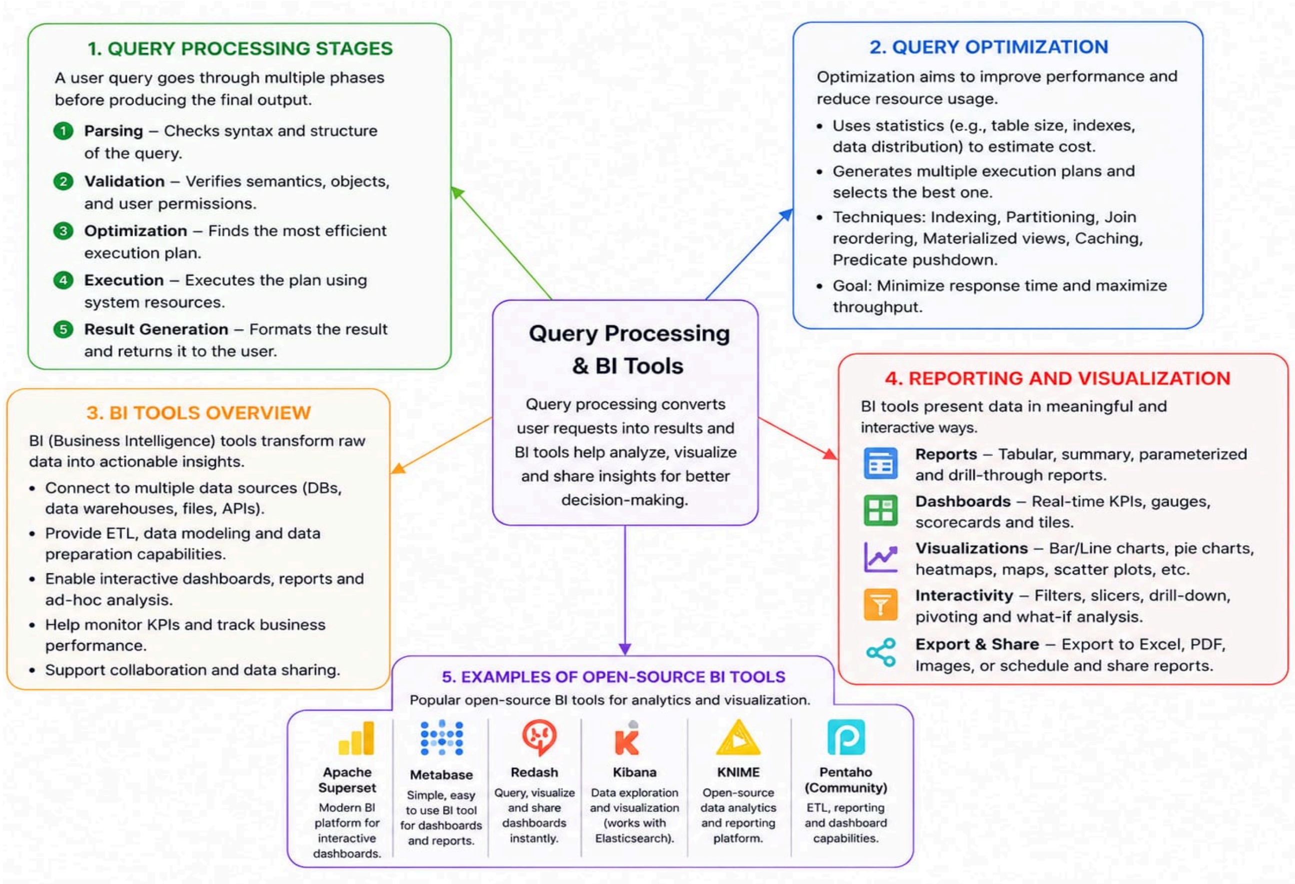
Query processing stages

Query optimization

BI tools overview

Reporting and visualization

Examples of open-source BI tools



7: OLAP Application using KNIME

Task description

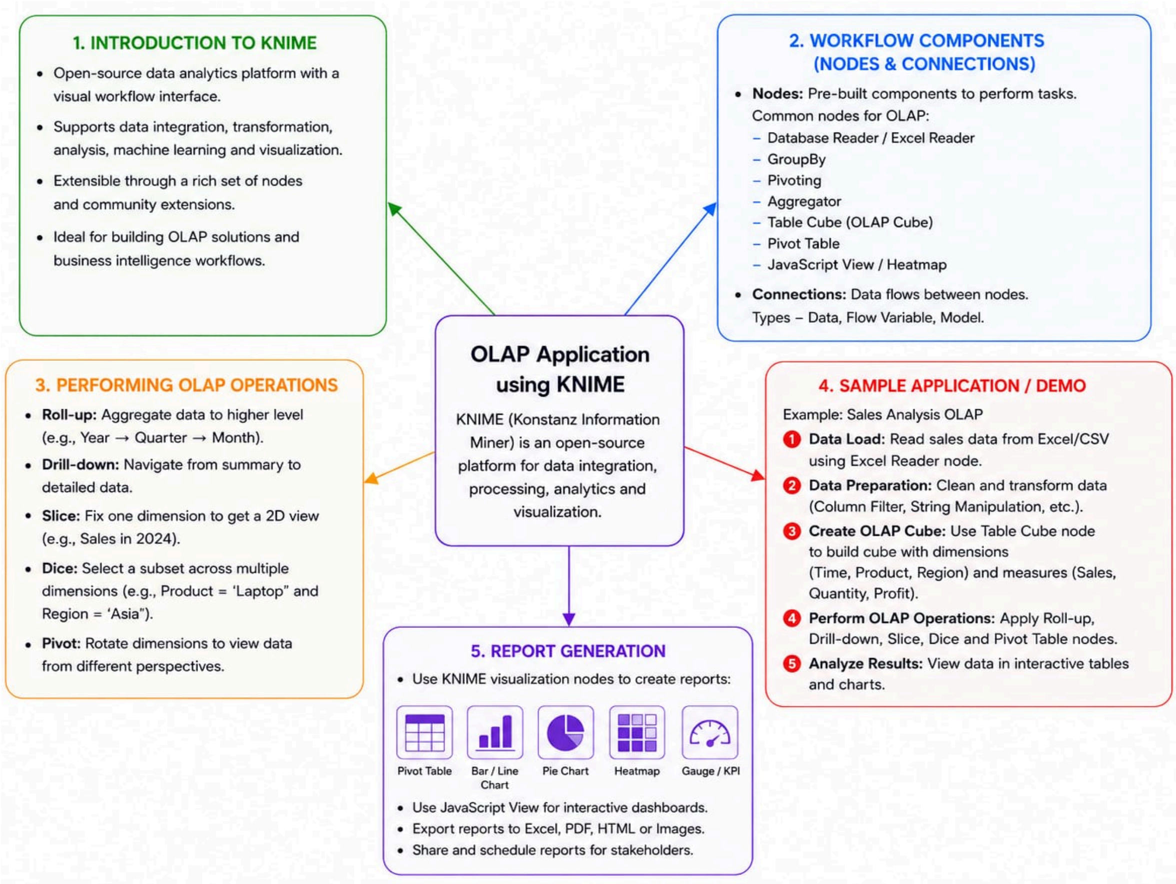
Introduction to KNIME

Workflow components (nodes, connections)

Performing OLAP operations

Sample application/demo

Report generation


OLAP Application using KNIME
KNIME (Konstanz Information Miner) is an open-source platform for data integration, processing, analytics and visualization.
1. INTRODUCTION TO KNIME
• Open-source data analytics platform with a visual workflow interface.
• Supports data integration, transformation, analysis, machine learning and visualization.
• Extensible through a rich set of nodes and community extensions.
• Ideal for building OLAP solutions and business intelligence workflows.
2. WORKFLOW COMPONENTS (NODES & CONNECTIONS)
• Nodes: Pre-built components to perform tasks. Common nodes for OLAP:
– Database Reader / Excel Reader
– GroupBy
– Pivoting
– Aggregator
– Table Cube (OLAP Cube)
– Pivot Table
– JavaScript View / Heatmap
• Connections: Data flows between nodes. Types – Data, Flow Variable, Model.
3. PERFORMING OLAP OPERATIONS
• Roll-up: Aggregate data to higher level (e.g., Year → Quarter → Month).
• Drill-down: Navigate from summary to detailed data.
• Slice: Fix one dimension to get a 2D view (e.g., Sales in 2024).
• Dice: Select a subset across multiple dimensions (e.g., Product = 'Laptop" and Region = 'Asia").
• Pivot: Rotate dimensions to view data from different perspectives.
4. SAMPLE APPLICATION / DEMO
Example: Sales Analysis OLAP
1 Data Load: Read sales data from Excel/CSV using Excel Reader node.
2 Data Preparation: Clean and transform data (Column Filter, String Manipulation, etc.).
3 Create OLAP Cube: Use Table Cube node to build cube with dimensions (Time, Product, Region) and measures (Sales, Quantity, Profit).
4 Perform OLAP Operations: Apply Roll-up, Drill-down, Slice, Dice and Pivot Table nodes.
5 Analyze Results: View data in interactive tables and charts.
5. REPORT GENERATION
• Use KNIME visualization nodes to create reports:
Pivot Table
Bar / Line Chart
Pie Chart
Heatmap
Gauge / KPI
• Use JavaScript View for interactive dashboards.
• Export reports to Excel, PDF, HTML or Images.
• Share and schedule reports for stakeholders.